

HOJEONG LEE

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I design assistive hardware and control systems that extend human hand capabilities. My research focuses on wearable haptic gloves, soft actuators, and wearable ergonomics to improve interaction between users and digital environments. I am currently developing electrostatic soft actuators that deliver vibrotactile and kinesthetic feedback, along with lightweight wearable gloves and systems for immersive tactile interaction in virtual environments.

EDUCATION

Ph.D. Candidate, Interdisciplinary Program in Robotics, School of Electrical Engineering
Korea Advanced Institute of Science and Technology (KAIST) | Daejeon, Republic of Korea | Aug 2025-Present
Advisor: Prof. Sang Ho Yoon

- Developing electrostatic haptic actuators, including micrometer-thin vibrotactile actuators and clutch actuators for force feedback; integrating embedded sensing into wearable gloves and evaluating interaction experiences in virtual environments.
- Completed coursework in robot kinematics, human–robot interaction, teleoperation and system control, and sensors and instrumentation to support robust haptic system design for telepresence and teleoperation.

M.S. in Graduate School of Culture Technology
Korea Advanced Institute of Science and Technology (KAIST) | Daejeon, Republic of Korea | Aug 2023-Aug 2025
Advisor: Prof. Sang Ho Yoon

- **Thesis: "Wearable Haptic Interface for Shape Perception through Spatiotemporal Vibrotactile Patterns and Kinesthetic Feedback"** Committee: Sang Ho Yoon, Jeongmi Lee, and Huitaek Yun.
- Developed a 1 DoF haptic glove using an electrostatic clutch and haptic rendering algorithms, then evaluated user experience in virtual environments.
- Developed a 1 DoF tendon-driven haptic glove with vibrotactile actuators and a haptic rendering system for reproducing object shapes; evaluated perception and user experience in virtual environments.
- Completed coursework in augmented reality, human–computer interaction, artificial intelligence and machine learning, and ergonomics.

B.F.A. in Industrial Design, Department of Design
Seoul National University | Seoul, Republic of Korea | Mar 2018-Aug 2023 | Graduation Advisor: Prof. Eui-Chul Jung

PUBLICATIONS

Journal Articles

1. A Novel Flexible Shaft-Driven Mechanism for Multidirectional Active Force Feedback in Haptic Gloves: A Preliminary Study

*Daegeun Park, **Hojeong Lee**, Sang Ho Yoon, Sung-Uk Jung*
IEEE Robotics and Automation Letters, 11(10), 11118–11125 (Aug 12, 2026). SCIE, IF 5.3.
https://doi.org/10.1109/LRA.2026.3723322

- Proposed and validated a flexible-shaft-driven haptic glove mechanism capable of providing multidirectional active force feedback during natural finger motion. Designed an exoskeleton with two linear actuators and two flexible shafts per finger. Simulated trajectories achieved mean directional errors of ≤6.9° and positional errors of ≤3.6 mm during combined motion;

force output errors were ≤8.6% relative to target forces in two finger postures. Users identified eight force-feedback directions with 76.0% average accuracy.

2. VibGrasp: Spatiotemporal Vibration Based Multimodal Haptic Rendering with a Lightweight Exo-Glove for 3D Shape Perception

***Hojeong Lee**, Eunho Kim, Rachel Kim, Sang Ho Yoon*
IEEE Transactions on Haptics, 19(3), 566-577 (Apr 20, 2026). SCIE, IF 3.1.

https://doi.org/10.1109/TOH.2026.3685691

- Developed a lightweight exoskeleton haptic glove and rendering method to improve grasping interactions in virtual environments. Verified comfortable wearability and a 77.8% feedback transmission rate to intended finger locations. Reduced tendon slack with a torsion-spring spool, and used vibrotactile funneling to stimulate multiple finger locations with only two actuators. Developed shape-specific spatiotemporal patterns and a virtual environment; user studies found significant improvements in shape discrimination and perceived realism over prior approaches.

3. EStatiG: Wearable Haptic Feedback with Multi-Phalanx Electrostatic Brake for Enhanced Object Perception in VR

Nicha Vanichvoranun, **Hojeong Lee***, Seoyeon Kim, Sang Ho Yoon (*First Co-authors)*
Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT), 8(3), Article 131 (Sep 9, 2024). IF 4.5, Q1.

https://dl.acm.org/doi/10.1145/3678567

- Developed a compact, compliant electrostatic clutch and multi-phalangeal haptic glove to improve wearability relative to bulky rigid systems. Fabricated lightweight actuators from TPU, PVDF film, and conductive fabric, and increased contact area and output efficiency using multilayer stacking. Actuators delivered an average force of 11 N per phalanx at 350 VAC. Designed and conducted virtual-object grasping studies, demonstrating benefits for user experience and haptic feedback delivery.

Conference Papers

4. 3D Shape Perception through Spatiotemporal Vibrotactile Patterns with Kinesthetic Feedback

***Hojeong Lee,** Eunho Kim, Sang Ho Yoon*
IEEE World Haptics Conference (WHC), July 2025. Selected as an outstanding paper (top 10%) and invited for journal publication.

5. Optimizing Button Size to Enhance User Experience for Augmented Reality Interactions in Mobile Scenarios

***Hojeong Lee,** Yeahyoung Jeon, Keisuke Kudo, Paul Ssemakula, Seonghyeok Park, Shuping Xiong*
22nd Triennial Congress of the International Ergonomics Association (IEA), Aug 2024.

6. ThermicVib: Enabling Dynamic Thermal Sensation with Multimodal Haptic Glove for Thermal-Responsive Interaction

*Hyungil Yi, **Hojeong Lee,** Sang Ho Yoon*
IEEE International Symposium on Mixed and Augmented Reality (ISMAR), Oct 2024.

Posters and Demonstrations

7. FlexVib: Multimodal Haptic Rendering of Object Shapes with Independent Finger Kinesthesia and Spatiotemporal Vibration

***Hojeong Lee,** Daegeun Park, Sung-Uk Jung, Sang Ho Yoon*
IEEE AsiaHaptics Conference 2026 (accepted).

8. Effect of Movement Velocity and Flexion Angle on Realistic Haptic Reproduction

Dongkyu Kwak, **Hojeong Lee***, Sang Ho Yoon (*Equal first authors)*
IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR Adjunct), Oct 2025.

9. Adaptive and Immersive XR Interactions with Wearable Interfaces (Demo of KAIST HCI Tech Lab)

Sang Ho Yoon, Youjin Sung, Kun Woo Song, Kyungeun Jung, Kyungjin Seo, Jina Kim, Hyungil Yi, Nicha Vanichvoranun, Hanseok Jeong, ***Hojeong Lee***

CHI Conference on Human Factors in Computing Systems (CHI Interactivity), May 2024. People's Choice Best Demo Award.

AWARDS

Outstanding Research Award (Dean´s Award)

College of Liberal Arts and Convergence Science, KAIST | 2024

Popular Choice Winner, CHI 2024 Interactivity *| 2024*

RESEARCH PROJECTS

Development of Spatial Media Technology and Interaction Technology for Convergence of the Real and Virtual World (High-Speed Multimodal 3D Haptic Rendering)

Electronics and Telecommunications Research Institute (ETRI) | Jan 2024-Present

- Contributed to development of haptic gloves and rendering methods for effective tactile feedback during remote interaction in virtual environments. Participated throughout the development of haptic glove using LRA and flexible-shaft, providing vibration and active force feedback for 1 and 2 DOF finger motion [1]. Presented and demonstrated the work as "XR-Based Telepresence Enhancement Technology" at the 2025 ETRI Conference.

Development of a Low-Cost Eyeglass-Type Eye-Tracking Platform Using an Embedded Eye-Tracking Module

KAIST | Jun 2026-Present

- Contributed hardware design to an eye-tracking glasses commercialization project, aimed at improving usability and accessibility over expensive, difficult-to-customize commercial systems for gaze-based interaction.

Emotional Expression through Robot Arm Movement

XYZ Co., Ltd. (Collaborative Research) | Aug 2025-Present

- Contributed ideation and experimental design to a motion-authoring system that enables non-roboticist motion designers to create emotionally expressive robot gestures from human movement videos.

Spatiotemporal Vibration with Kinesthetic Feedback for 3D Shape Representation

2025-2026 | Project Lead

- Developed an active haptic system combining force and vibration patterns to convey object shapes, and evaluated user perception and virtual-environment experience [2, 4]. Led device hardware development, control PCB design, haptic rendering in C, virtual-environment implementation in Unity (C#), and perceptual/user-experience study design and analysis.

Multi-Phalangeal Electrostatic Clutch for Object Shape Perception in VR

2023-2024 | Co-Lead

- Developed a multiphalangeal haptic glove using electrostatic actuato,; analyzed force-feedback perception and user experience [3]. Co-led clutch development, Unity (C#) virtual environments, and perception/user-experience study design.

Evaluating Priority Elements for Realistic Haptic Reproduction

2025 | Co-Lead

- Conducted a perception study to investigate the relative importance of movement velocity versus motion amplitude in realistic hand-motion haptic reproduction [8]. Co-led haptic rendering system development (C) and study design.

A Wearable Multimodal Sensing Framework for Adaptive Interaction in Extended Reality

National Research Foundation of Korea (NRF) | Aug 2023-Aug 2025

Developing the Model and Criteria for Evaluating Motion Sickness in Automobile Driving Scenarios

Hyundai NGV | Jan-Jun 2023 | Research Assistant, Human Interface System Laboratory, Seoul National University (Advisor: Prof. Myung Hwan Yun)

- Conducted a literature review of factors contributing to motion sickness in autonomous-driving contexts; assisted with biosignal acquisition, quantitative Likert-scale surveys, and qualitative interviews during on-road motion-sickness studies.

INTERNATIONAL EXPERIENCE

Elite Summer School of Robotics and Entrepreneurship

University of Southern Denmark | Odense, Denmark | Aug 2026

- Completed a collaborative project with LEGO developing a vision-based part-classification system.

International Summer School

Frankfurt University of Applied Sciences | Frankfurt, Germany | Jul 2019

- Worked on the "Urban Redesign Project for Frankfurt Neue Strasse and Offenbach Station," developing concepts for urban space redesign and public amenities.

EXPERIENCE & ACADEMIC SERVICE

President, Student Council, Department of Design

Seoul National University | Dec 2019-Dec 2020

- Organized departmental events through virtual meetings during the COVID-19 pandemic and coordinated crowdfunding-based sales of student design work.

Student Volunteer

UbiComp / ISWC 2024

Peer Reviewer

CHI Late-Breaking Work 2024; Augmented Humans 2025; WHC 2025; ISMAR 2025; UIST 2025; ISMAR Posters 2025; CHI 2026; HRI 2026; CHI Posters 2026; EuroHaptics 2026; DIS 2026.

Invited Talk

- **SNUD Design Week**
  Seoul National University | Dec 2025
- **Korean Haptics Society**
  Aug 2024 | Presentation: "Development of a Multi-Joint Electrostatic Brake-Based Wearable Haptic Device for Object Perception in VR."

EXHIBITIONS

Seoul Design 2022 – Dream Trade

Sponsored by the Seoul Design Foundation | Oct 2022 | Collaborators: Haeun Ki, Yuri Kim, Heejung Yoo, Hojeong Lee, Seyun Jun, Sung Yun Jang

ZER01NE DAY 2021 – headspace

Sponsored by Hyundai Motor Group | Nov 2021 | With Geunwook Kim

- Contributed wearable-design concept development, prototyping, and design assistance.

TECHNICAL SKILLS

Fabrication: Electrostatic/electroadhesive actuators; Polymer (silicone) fabrication; Machining; CNC; 3D printing (FDM, SLA).

Modeling: Rhino 3D; Grasshopper; CAD; ANSYS.

Analysis: Force transmission; Movement accuracy (position and orientation); Human perception; User experience (qualitative and quantitative)

Programming: C; C# (Unity); Java; Python.